1 The energy E stored in a certain electronic component is given by

$$E = \frac{Q^n}{2k}$$

where Q is the total charge in the component, n is an unknown integer and k is a physical quantity with SI base units $A^2\,s^4\,kg^{-1}\,m^{-2}$.

What is the value of n?

A -2 **B** -1 **C** 1 **D** 2

Ans: D

$$\text{Units of } E = \frac{(\text{Units of } Q)^n}{\text{Units of } k}$$

$$kg\ m^2\ s^{-2} = \frac{A^n\ s^n}{A^2\ s^4\ kg^{-1}\ m^{-2}}$$

$$n = 2$$